



COLLEGE OF
APPLIED BIOLOGISTS

Credentialing Standard

Effective June 2025

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1.0 The College of Applied Biologists

The College of Applied Biologists (the College) is the regulatory body for applied biology professionals in British Columbia (BC) who work in the natural resource sector. The College has been granted authority from the province of BC to register applied biology professionals, grant licenses and protected titles and regulate its registrants under the [Professional Governance Act](#), [Professional Governance Regulations](#), [Applied Biologists Regulation](#) and the [College Bylaws](#).

2.0 Registration

This document is specific to the “Apply to Join” and “Reclassify” applications for registration.

Details on registration streams and associated requirements are provided below in section 5.0 and in each registrant category section of this document (Sections 6.0 – 9.0).

Registration with the College requires the following:

1. A completed application that meets the College’s requirements for the registrant category applied for (e.g., Applied Biology Technician)
2. Completion of all mandatory training, and
3. Paying all applicable fees (e.g., annual dues and seal)

The Registrar may approve an application for registration under Part 5 of the [College Bylaws](#).

For details on how applications are assessed please refer to [Policy 5 – 200 Registration Application Assessment](#).

3.0 Registration Applications for Apply to Join or Reclassify

The application, assessment and registration process is completed online through the College’s portal <https://portal.cab-bc.org/applicants/start>.

Tables 1.0 and 2.0 below summarize the application requirements for practicing registrants and in Training registrants.

Practicing Registrants – are registrants who have met all the College’s education/training and experience requirements in one of the following registrant categories:

- Registered Professional Biologist (RPBio)
- Registered Biology Technologist (RBTech)
- Applied Biologist Limited License (AB-LL)
- Applied Biology Technician (ABT)

Table 1.0 Summary of the requirements for the RPBio, RBTech, ABT and AB-LL. application for registration with the College.

Practicing Registrant Category	RPBio	RBTech	AB-LL.	ABT
Proof of Education/Training	Yes	Yes	Yes	Yes
Experience	Yes	Yes	Yes	Yes
Professional Practice Competencies	Yes	Yes	Yes	Yes
Professional Work Products	Yes	Yes	Yes	No
References	Yes	Yes	Yes	Yes
Indictable Offence Declaration	Yes	Yes	Yes	Yes
Application Fee	Yes	Yes	Yes	Yes

In Training Registrants – are registrants who have meet all the College’s education requirements and are working towards fulfilling the experience requirements in one of the following registrant categories:

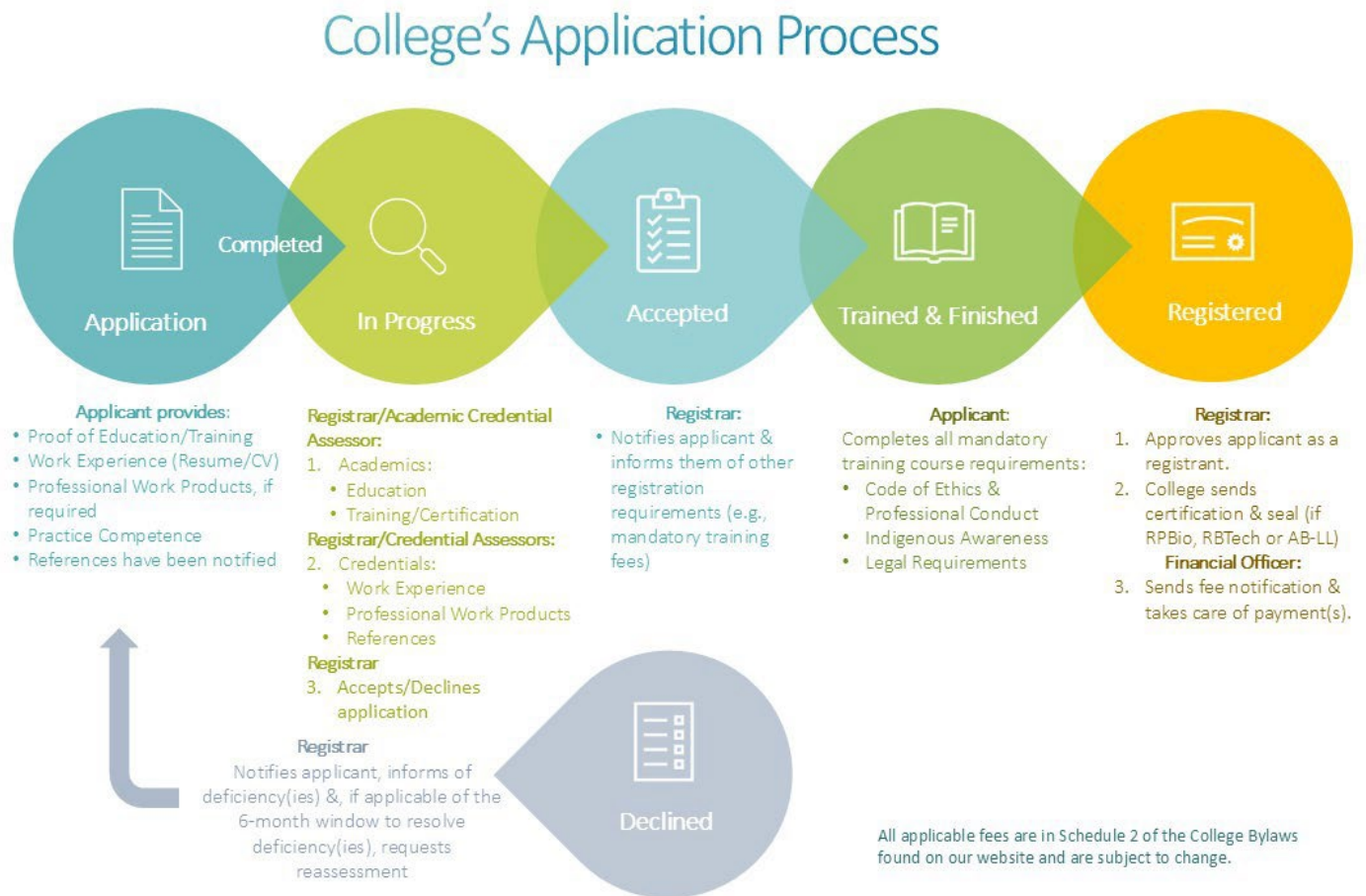
- Biologist In training (BIT)
- Registered biology technologist in training (RBTech in Training)
- Applied biology technician in training (ABT in Training)

Table 2.0 Summary of the requirements for the BIT, RBTech in Training and ABT in Training application for registration with the College.

In Training Registrant Category	BIT	RBTech in Training	ABT in Training
Proof of Education/Training	Yes	Yes	Yes
Indictable Offence Declaration	Yes	Yes	Yes
Application Fee	Yes	Yes	Yes

Figures 1.0 and 2.0 outline the application and fee payment processes.

Figure 1.0 College's Application Process



A Completed application means the application fee has been paid and all documentation (e.g., transcripts, lead author declaration, references etc.) have been received. See [Understanding your application status](#) for details.

Figure 2.0 College's Fee Payment Schedule

Fees Payment Schedule



*See Schedule 2 of the [College Bylaws](#) for details of fees.

4.0 College Accredited Programs – Stream 1

Accredited programs have met the education requirements set by the College in this Credential Standard for the registrant category the program is accredited for. Applicants can only apply through Stream 1 if they have completed/graduated from a College Accredited program during the timeframe of the program's accreditation.

5.0 Credential Requirements for All Applications

5.1 Education/Training Requirements

Each registrant application category requires proof of education (e.g., official transcripts, certificate, on the job training). These are summarized below in Table 3.0. Details of specific education requirements (e.g., courses, training modules) for each registration category are provided in Sections 6.0 – 9.0..

Table 3.0 Summary of the proof of education required for each category for registration with the College.

Registrant Application Category	Proof of Education/Training Required			
	Stream 1	Stream 2	Stream 3	Stream 4
ABT	Official Transcripts/Certification		Not Applicable	Not Applicable
ABT in Training				
RBTech	Official Transcripts showing diploma(s) granted.		Official Transcripts	
RBTech in Training			Not Applicable	
RPBio	Official Transcripts showing degree(s) granted.		Official Transcripts	Official Transcripts

		showing degree(s) granted.	showing M.Sc. &/or PhD degree(s) granted in an applied biology topic.
Biologist in Training (BIT)		Not Applicable	
AB - LL	AB – LL (Steam 1)		
	Proof of education that supports the limited license area being applied for		

Note:

- An applicant with a diploma(s), degree(s) can apply for the ABT or ABT in Training category, if they meet the required course work/module requirements.
- An applicant with a degree(s) can apply for the RBTech or RBTech in Training category, if they meet the required course requirements.

5.2 Experience

All practicing registrant applicants are required to demonstrate that they meet the experience requirements for the registrant category they have applied for. Experience must fall within the definition of the “practice of applied biology” in the [Applied Biologists Regulation](#), which is the provision of

- advice or services that
 - are based on biological sciences, and
 - relate to aquatic or terrestrial ecosystems or the living organisms, habitats or processes of those ecosystems, or
- advice or services that are ancillary to those described in paragraph (a).

5.2.1 Volunteer Experience

Applicants can use volunteer experience towards meeting the experience requirement, if it falls within the practice of applied biology as defined in the [Applied Biologists Regulation](#).

5.2.2 Graduate Degree Experience

Applicants with a thesis-based Master of Science degree in an applied biology topic can claim 12 months to meet the experience requirement.

Applicants with a Doctor of Philosophy – (PhD) degree in an applied biology topic can claim 24 months to meet the experience requirement.

5.2.3 Registrant Experience

An applicant who is already a registrant with the College (or an equivalent professional regulator in another jurisdiction) may claim all their applied biology experience gained while registered (pre- or post-graduation).

5.3 Professional Practice Competencies

All practicing registrant applicants are required to demonstrate that they meet 1 of the three key indicators of all 7 of the College’s Professional Practice Competencies (Appendix A). The College’s

Professional Practice Competencies are broadly encompassing and not limited to applied biology to represent all the functions of a professional and can include those specific to an organization or employer.

5.4 Professional Work Products

Applicants for the following registrant categories are required to submit Professional Work Products (see Section 13.0 for definition and examples of what the College accepts as a PWP) as part of their application:

- Registered Professional Biologist (RPBio)
- Registered Biology Technologist (RBTech)
- Applied Biologist Limited License (AB-LL)

Where a report has multiple authors, the role of the applicant must be corroborated by the lead author. A lead author is sent an email directly from the online application and they receive a link directly to the lead author section in the applicant's application. They cannot see or have access to any other portion of the application.

See specific registrant application requirements in Sections 6.0 – 9.0.

5.5 Reference Requirements

A reference needs to:

- Confirm the applicant demonstrates the Professional Practice competencies defined by the College
- Be knowledgeable of the applicant's ability and work as an applied biology professional (i.e., they must be someone the applicant has worked with), and
- Not be related to the applicant unless the relative is the supervising professional.

Note: The College recognizes the variety of experience and employer(s) an applicant may have. It is not required that references be registrants of the College or be from different employers. Applicants are encouraged to include references from more than one employer when possible.

A reference is sent an email directly from the online application. The reference(s) receives a link directly to the reference section in the applicant's application. They cannot see or have access to any other portion of the application.

5.6 Mandatory Training

All applicants who have been approved are required to complete the College's mandatory training as set out by the [College Bylaws](#).

6.0 Applied Biology Technician (ABT) and ABT in Training Application Requirements

There are three streams to registration in the ABT category (Figure 3.0). Streams 1 and 2 require education and a minimum of 1 year of experience in applied biology in the 10 years prior to application. Stream 3 requires 3 years of experience in applied biology in the 10 years prior to application. Table 4.0 summarizes the requirements for each stream.

Figure 3.0 Streams of Entrance for the Applied Biology Technician Registrant Category.

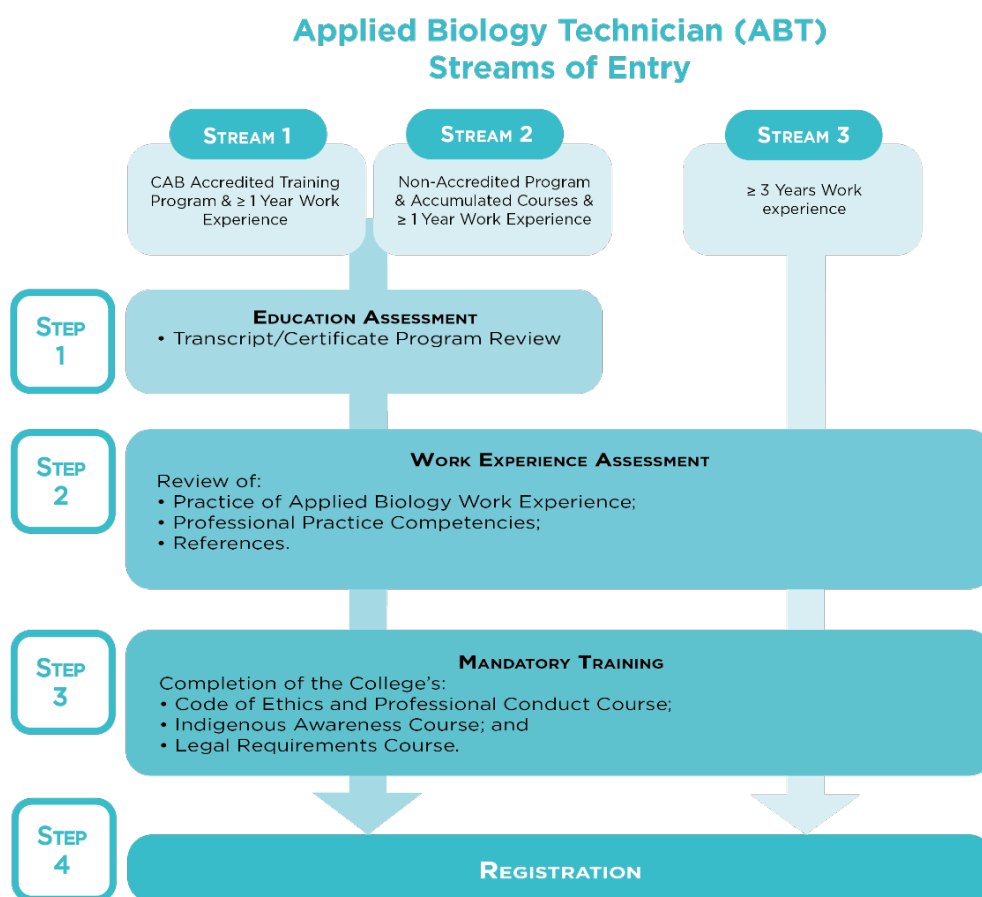


Table 4.0 provides the requirements for ABT and ABT in Training applicants for all 3 streams. An ABT in Training application only needs to meet the academic requirement and will work on other requirements (e.g., experience) during their training period.

Table 4.0 A Summary of the ABT and ABT in Training Application Requirements

Requirements	ABT	ABT in Training	ABT
	Streams 1 & 2		Stream 3
Academics	A minimum of 200 hours or 5 instructional weeks of course work, with or without a certificate, which includes courses or modules in each of the following areas: • Natural Resource Survey Skills • Applied Biology Identification Skills • Field Navigation		Not Applicable
Experience	≥ 1 year of applied biology experience within the 10 years prior to application.	Not Applicable	≥ 3 years of applied biology experience within the 10 years prior to application.

Professional Practice Competencies	Demonstrate that you meet all the College's Professional Practice Competencies.		Demonstrate that you meet all the College's Professional Practice Competencies.
References	1		2
Complete the College's Mandatory Training Courses	• Code of Ethics & Professional Conduct • Indigenous Awareness • Legal Requirements		

6.1 Proof of Education

The applicant must submit official transcripts or a certificate demonstrating successful completion of a training course (diploma, degree(s)) from a recognized training or post-secondary institution which includes the course or course modules required as set out above.

6.2 Experience Requirements

Experience must align with the definition of the “practice of applied biology” as defined in the [Applied Biologists Regulation](#).

- Streams 1 and 2 - The applicant must demonstrate a minimum of 1 year of work experience practicing applied biology within the 10 years prior to application.
- Stream 3 – The applicant must demonstrate a minimum of 3 years’ experience practicing applied biology within the 10 years prior to application.

Additional information on experience requirements is set out in section 5.0.

6.3 Professional Practice Competencies

Applicants of all streams must demonstrate how they meet 1 of the 3 key indicators of each of the [College's Professional Practice Competencies](#).

7.0 Registered Biology Technologist (RBTech) and RBTech in Training Application Requirements

There are 3 streams to register as an RBTech:

- Streams 1 and 2 require a diploma, courses in 5 areas (Table 5.0), and a minimum of 2 years' experience in applied biology in the 10 years prior to application.
- Stream 3 requires 20 post-secondary courses including courses in 5 areas (Table 5.0) and a minimum of 3 years' experience in applied biology in the 10 years prior to application.

Figure 4.0 illustrates the streams, and Table 5.0 summarizes the requirements for RBTech applications.

Figure 4.0 Streams of Entrance for the Registered Biology Technologist Registrant Category.

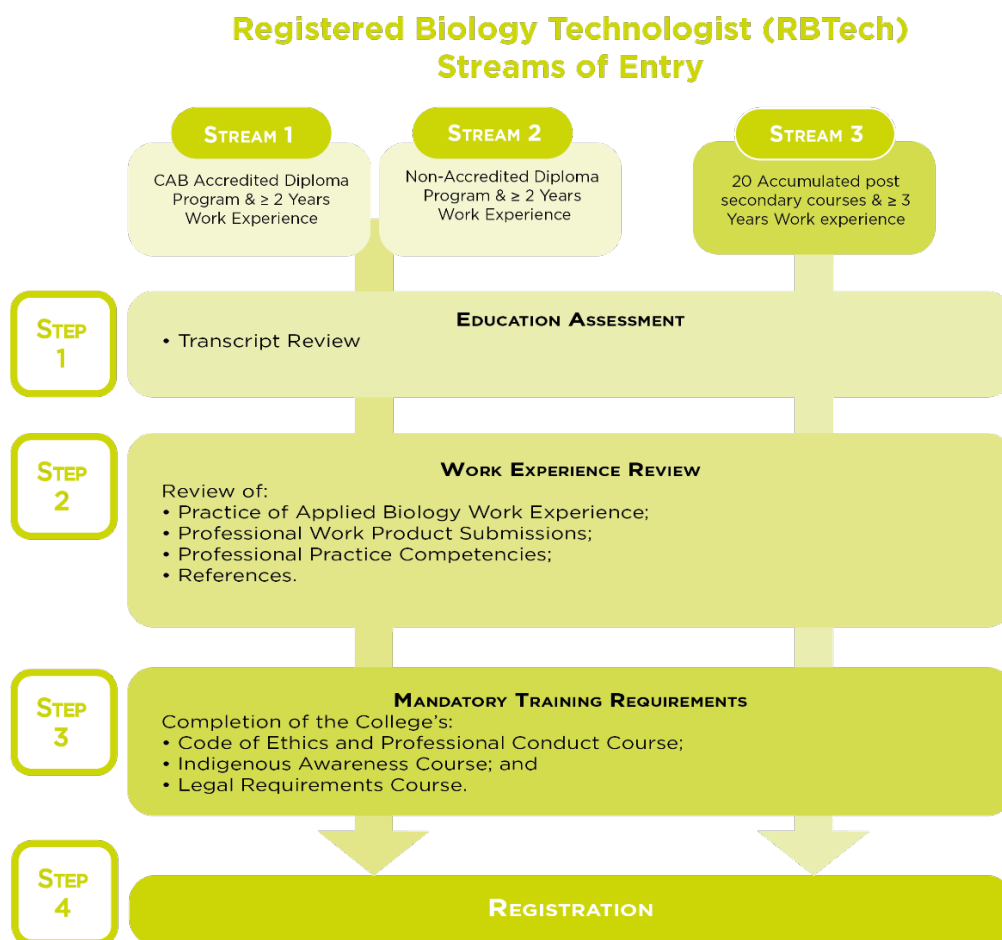


Table 5.0 Summary of the RBTech and RBTech in Training Application Requirements

Requirements	RBTech	RBTech in Training	RBTech
	Streams 1 & 2		Stream 3
Education/Training	• Diploma • Courses in each of the following categories: • Applied Biology • Communications • Environmental Science • Field/Laboratory Techniques • Numeracy (Mathematics/Statistics)		Minimum of 20 courses, including courses in each of the following categories: • Applied Biology • Communications • Environmental Science • Field/Laboratory Techniques • Numeracy (Mathematics/Statistics)
Experience	≥ 2 years of applied biology experience within the 10 years prior to application.	Not Applicable	≥ 3 years of applied biology experience within the 10 years prior to application.
Professional Practice Competencies Refer to Section	Demonstrate that you meet all the College’s Professional Practice Competencies.		Demonstrate that you meet all the College’s Professional Practice Competencies.
Professional Work Products (PWPs) (see below)	2 to 4 PWPs that meets the criteria below		2 to 4 PWPs that meets the criteria below
References	2		2
Complete the College’s Mandatory Training Courses	• Code of Ethics & Professional Conduct • Indigenous Awareness • Legal Requirements		

7.1 Proof of Education

The applicant must submit official transcripts demonstrating successful completion of a diploma (or degree(s)) or 20 completed courses from a recognized post-secondary institution which includes the course requirements as set out above.

7.2 Experience Requirements

Experience must align with the definition of the “practice of applied biology” as defined in the [Applied Biologists Regulation](#).

- Streams 1 and 2 - The applicant must demonstrate a minimum of 2 years (24 months) experience practicing applied biology within the 10 years prior to application. Up to 1 year (12 months) of this experience may be pre-diploma.
- Stream 3 – The applicant must demonstrate a minimum of 3 years (36 months) experience practicing applied biology within the 10 years prior to application.

Additional information on experience requirements is set out in section 5.0.

7.3 Professional Practice Competencies

Applicants of all streams must demonstrate how they meet 1 of the 3 key indicators of each of the [College's Professional Practice Competencies](#).

7.4 Professional Work Products Requirements

Intent: The professional work product requirement is important to the RBTech application as it is a mechanism for applicants to demonstrate a combination of their independent thought as an applied biology professional, and how their practice in applied biology and professional competence align with the [College's Professional Practice Competencies](#).

In their application, an applicant must:

1. Provide a minimum of 2 and up to a maximum of 4 separate professional work products in applied biology that, between the provided products, includes the following:
 - Method(s)
 - Data analysis within established standards, parameters, or guidelines
 - Conclusion(s) and/or Recommendations, and
 - Communicates technical scientific information to a non-expert audience.
2. Explain how, between the professional work products, they meet at least one key indicator in each the following professional competencies:
 - Scientific Knowledge & Concepts
 - Laws, regulations & policy
 - Standards & Practices, and
 - Communication.
3. Describe the purpose/intent/background of each of the professional work product(s) submitted.

8.0 Registered Professional Biologist (RPBio) and Biologist in Training (BIT) Application Requirements

There are 4 registration streams in the RPBio category:

- Streams 1 and 2 require a bachelor's or a master's degree with 25 science courses, 13 in biology and several required courses in a variety of areas, and a minimum of 3 years' experience in applied biology in the 10 years prior to application.
- Stream 3 requires a bachelor's or a master's degree with 25 science courses, 13 in biology, and a minimum of 15 years' experience in the practice of applied biology.
- Stream 4 requires either a Master of Science degree or a Doctor of Philosophy degree in an applied biology topic, 25 science courses, 13 in biology, and a minimum of 13 or 11 years' experience in the practice of applied biology (MSc or PhD, respectively).

Figure 5.0 illustrates the streams, and Table 6.0 summarizes the requirements for an RPBio application.

Figure 5.0 Streams of Entrance for the Registered Professional Biologist Registrant Category.

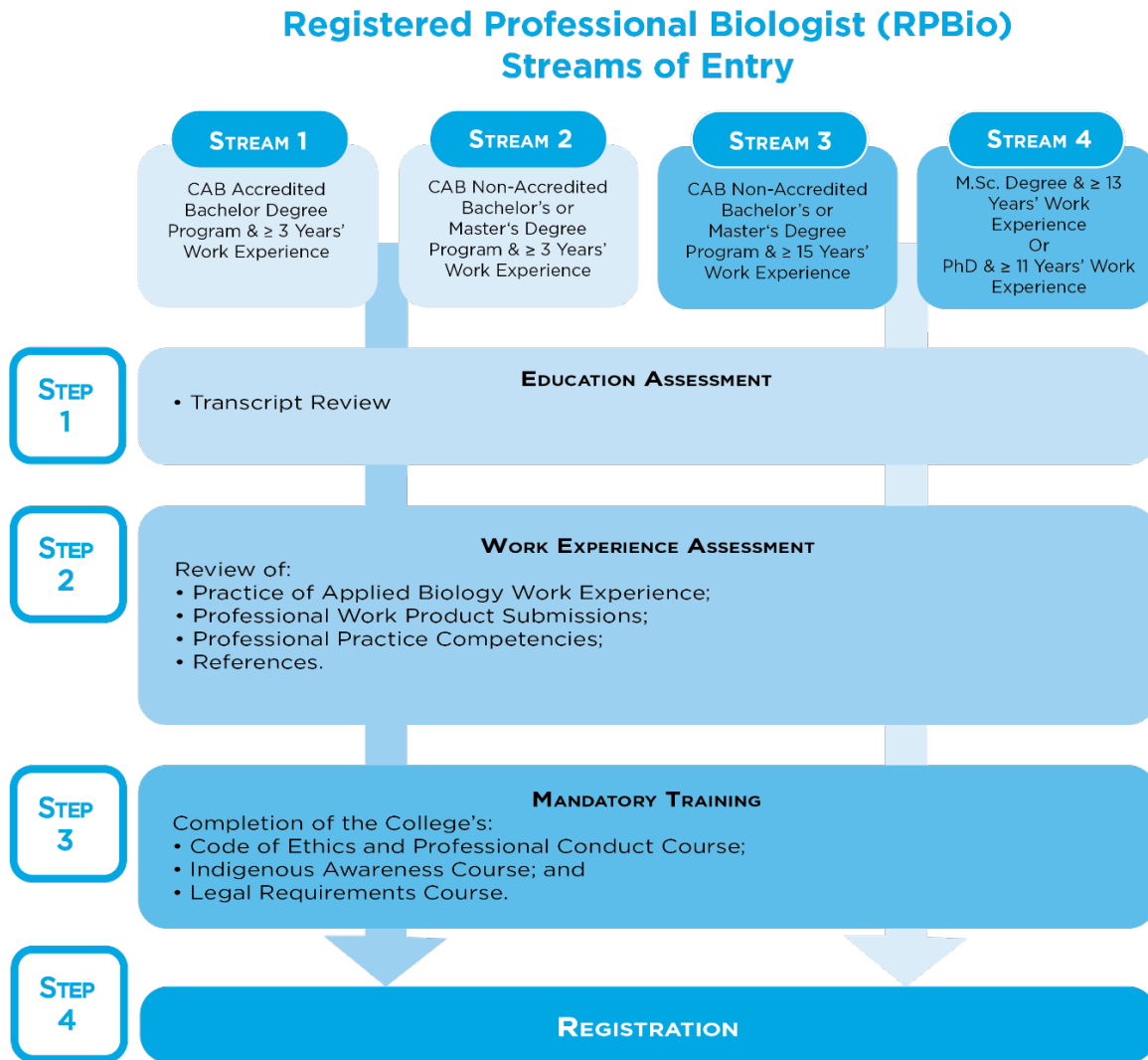


Table 6.0 A Summary of the RPBio and Biologist in Training Application Requirements

Requirements	RPBio	Biologist in Training (BIT)	RPBio	
	Streams 1 & 2		Stream 3	Stream 4
Academics	- Bachelor’s or Master’s Degree 25 Science Courses, 13 in biology - Core courses in the following: 1st year or higher - Communications - Chemistry - Numeracy (Mathematics/Statistics) 2nd year or higher - Applied Biology - Ecology - Numeracy must be Statistics 3 of the following 2nd year or higher courses: - Cell Biology - Genetics - Physiology - Systematics - Evolution		Bachelor’s or Master’s Degree with 25 Science Courses, 13 in biology.	Master of Science degree/PhD in an applied biology topic, with 25 Science Courses & 13 in biology.
Experience	≥ 3 years of applied biology experience within the 10 years prior to application	Not applicable	≥ 15 years of applied biology experience	- Master of Science degree in an applied biology topic plus ≥ 13 years of applied biology experience - PhD in an applied biology topic plus ≥ 11 years of applied biology experience
Professional Practice Competencies Refer to Section	Demonstrate that you meet all the College’s Professional Practice Competencies.		Demonstrate that you meet all the College’s Professional Practice Competencies.	
Professional Work Products (PWPs)	2 to 4 PWPs that meets the criteria below		2 to 4 PWPs that meets the criteria below	
References	3		3	3
Complete the College’s Mandatory Training Courses	- Code of Ethics & Professional Conduct - Indigenous Awareness - Legal Requirements			

8.1 Proof of Education

The applicant must submit official transcripts demonstrating successful completion of an undergraduate or graduate degree from a recognized post-secondary institution or in another biological science which includes 25 science courses, 13 of which must be in biology plus additional course requirements as set out above in Table 6.0.

8.2 Experience Requirements

Experience must align with the definition of the “practice of applied biology” as defined in the [Applied Biologists Regulation](#).

- Streams 1 and 2 - The applicant must demonstrate a minimum of 3 years (36 months) experience practicing applied biology within the 10 years prior to application. Up to one year (12 months) of this experience may be pre-degree.
- Stream 3 – Bachelor’s or Master’s degree and ≥ 15 years (180 months) of applied biology experience. Up to one year (12 months) of this experience may be pre-degree.
- Stream 4 – Either:
 - a Master of Science degree in an applied biology topic and ≥ 13 years (156 months) of applied biology experience, OR
 - a PhD degree in an applied biology topic plus ≥ 11 years (132 months) of applied biology experience.Up to one year (12 months) of this experience may be pre-undergraduate degree.

Additional information on experience requirements is set out in section 5.0.

8.3 Professional Practice Competencies

Applicants of all streams must demonstrate how they meet 1 of the 3 key indicators of each of the [College’s Professional Practice Competencies](#).

8.4 Professional Work Products Requirements for RPBio application

Intent: The professional work product (PWP) requirement is important to the RPBio application as it is a mechanism for applicants to demonstrate a combination of their independent thought as an applied biology professional, and how their practice in applied biology and professional competence align with the [College’s Professional Practice Competencies](#).

In their application, an applicant must:

1. Provide a minimum of 2 and up to a maximum of 4 separate professional work products in applied biology that, between the provided products, includes the following:
 - Method(s)
 - Data analysis
 - Conclusion(s) and/or Recommendations
 - Communicates technical scientific information to a non-expert audience.
2. Explain how, between the professional work products, they meet at least one key indicator in each the following professional competencies (see Appendix A for definitions):

- Scientific Knowledge & Concepts
- Laws, regulations & policy
- Standards & Practices
- Communication

3. Describe the purpose/intent/background of each of the work product(s) submitted.

9.0 Applied Biology – Limited Licensee (AB-LL.) Application Requirements

There is 1 stream of registration for the AB-LL. Registrant category. This category requires education/training in the area of the limited license being applied for, and five years of experience in applied biology, with a minimum of 2 years in the specific area of the Limited License application.

Figure 6.0 illustrates the registration process, and Table 7.0 summarizes the requirements for the AB-LL.

Figure 6.0 Stream of Entrance for the Applied Biology – Limited Licensee Registrant Category.

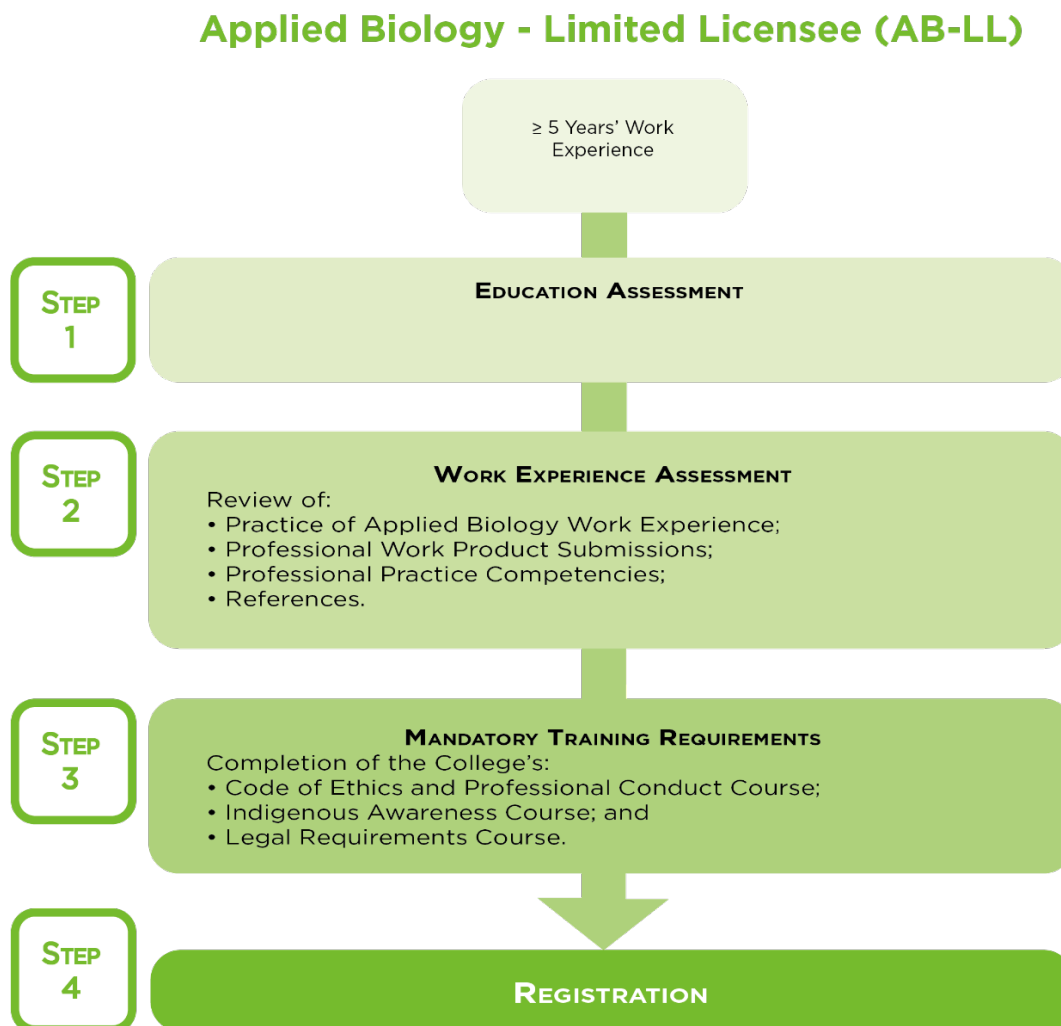


Table 7.0 Summary of the AB-LL Application Requirements

Requirements	AB-LL.
Education/Training	Proof of education that supports the license being applied for
Experience	≥ 5 years of applied biology experience with 2 years' experience specific to the area of the license application
Professional Practice Competencies	Demonstrate that you meet all the College's Professional Practice Competencies
Professional Work Products (PWP)	2 to 4 PWPs that meets the criteria below
References	3
Complete the College's Mandatory Training Courses	• Code of Ethics & Professional Conduct • Indigenous Awareness • Legal Requirements

9.1 Description of the specific area the limited license is for

Applicants must provide a description of the proposed specific area of the reserved practice the Limited License (LL) is for.

9.2 Proof of Education/Training

The applicant must submit official documents demonstrating successful completion of education/training in the specific area they are applying for. This could include official transcripts, certificate(s), diploma, degree(s).

9.3 Experience Requirements

5 years' experience in the practice of applied biology as defined in the [Applied Biologists Regulation](#) with 2 years' experience in the specific area of the Limited License application.

9.4 Professional Work Products Requirements

Intent: The professional work product requirement is important to the AB-LL application as it is a mechanism for applicants to demonstrate a combination of their independent thought as an applied biology professional in the area of their LL application, and how their practice in applied biology, specific to their LL application and professional competence align with the [College's Professional Practice Competencies](#).

In their application, an applicant must:

1. Provide a minimum of two and up to a maximum of four separate professional work products in the specific area of applied biology that the limited license application is for, between the provided products, includes the following:
 - Method(s)
 - Data analysis within established standards, parameters &/or guidelines
 - Conclusion(s) and/or Recommendations, and

- Communicates technical scientific information to a non-expert audience.
2. Explain how, between the professional work products, they meet at least one key indicator in each the following professional competencies:
 - Scientific Knowledge & Concepts
 - Laws, regulations & policy
 - Standards & Practices
 - Communication
 3. Describe the purpose/intent/background of each of the professional work product submitted.

10.0 Options to Resolve Education Deficiencies

An applicant who is deficient in meeting the specific course requirements may resolve the deficiency by:

- a) Challenging the course exam(s) through a recognized post-secondary institution
- b) Successfully completing the GRE examination in the area(s) in which they are deemed deficient
- c) Completing the missing course(s) at a recognized post-secondary institution, or
- d) Obtaining course credit from a post-secondary institution through Prior Learning Assessment and Recognition (PLAR) if available.

11.0 Steps Required After an Application is Approved

After an application has been approved, applicants must:

1. Pay for and complete the College's mandatory training courses, and
2. Pay fees (e.g., annual registration dues, seal fee).

Table 8.0 outlines the registrant categories with a seal fee applied. Annual dues are pro-rated to reflect the period during the College's fiscal year (January 1 to December 31) in which the application was approved.

Table 8.0 Summary of fee requirements for approved registrant categories.

Practicing Registrant Category	RPBio	RBTech	AB-LL	ABT
Mandatory Training Fee	Yes			
Registration Fee	Yes			
Seal Fee	Yes			Not Applicable

Schedule 2 – Fees and Charges of the [College Bylaws](#) sets out the current fees.

12.0 Definitions

Accredited program - has met the standard education requirements set by the College of Applied

Biologists for the registrant category the program is accredited for (e.g., RPBio, RBTech, ABT)

Communication - is written or presented to a non-expert audience (see definitions of 'non-expert audience' below).

Conclusion(s) - a summation of outcomes of data analysis, along with an interpretation of significance and implications.

Course - is defined as a one-semester course, which includes at least 3 lecture or lab hours per week

Data - facts, information, and/or statistics (e.g., measurements, metrics) collected for reference or analysis.

Data Analysis – the study and/or examination of data applying logical techniques to describe and illustrate, condense and recap, and evaluate data.

Methods – implementing a systematic or a particular form of procedure or process for accomplishing or approaching something such as, a task (e.g., information/data gathering, statistical analysis).

Non-expert audience – an individual or individuals who is not a subject matter expert(s) in the topic being communicated. This can include other applied biology professionals, clients, the public, other professionals.

Official transcript(s) – means a transcript(s) issued by the granting institution directly or via a third party (e.g., My eQuals, Mycreds, National Student Clearinghouse etc.) that includes a record of courses, marks, degrees, diploma or certificates or obtained from each university, college or training institute attended. Degree(s) granted or any foreign education earned at a post- secondary institution outside of Canada or the United States of America will be required to be evaluated by a credential evaluation service identified by the College (e.g., World Education Services (WES), International Credential Evaluation Service (ICES), or Comparative Education Service (CES)).

Professional Work Product(s) - includes but is not limited to a report, policy, assessment, plan, briefing note, memorandum, a review report (e.g., literature review) or scientific publication(s), thesis, dissertation, poster, presentation, email communication. Draft work products written by the applicant such as draft report, draft briefing note are accepted as long as they meet the intent. Draft work products are often provided to supervisors and others (e.g., clients, another professional signing off) for review and may be changed by supervisors, higher-level staff within the applicant's organization prior to being finalized. Bachelor's Honours Theses are accepted as a professional work product if they meet the communication competency¹. The College may accept other professional work products that meet the requirements.

Recognized - means the academic institution has met a quality assurance programs as defined by the British Columbia Council on Admission and Transfer (<https://www.bccat.ca/system/accreditation>) or similar definition in another jurisdiction.

Recommendation – an informed suggestion regarding the best course of action.

¹ [College's Professional Competencies](#)

APPENDIX A – Professional Practice Competencies and Competence Standard

PROFESSIONAL PRACTICE COMPETENCIES & COMPETENCE STANDARD

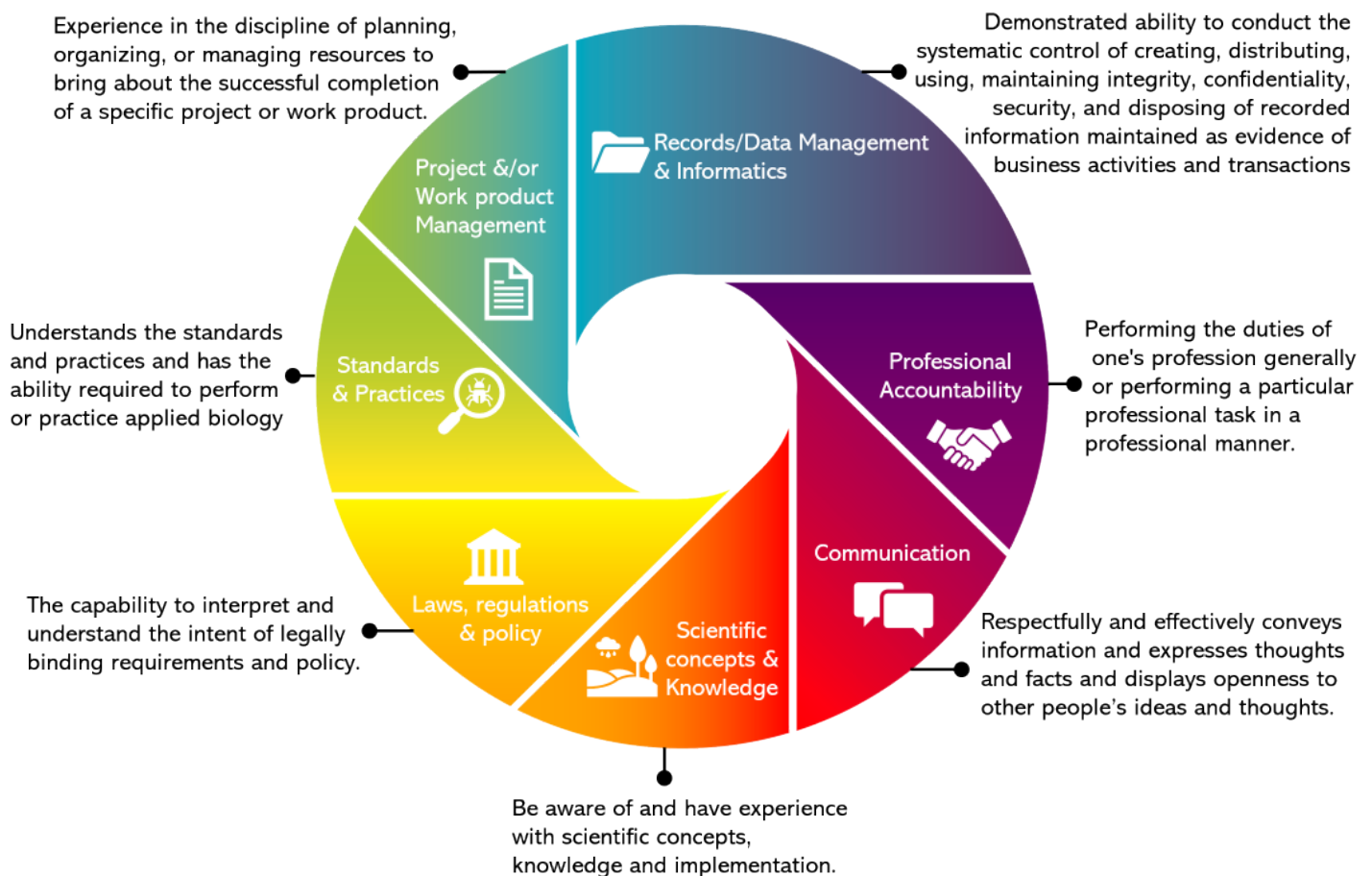
Fundamental principles of professional reliance are that all applied biology professionals meet high professional standards such as competence, expertise and work within their own area(s) of practice and level(s) of competency. A competency is more comprehensive than skill set(s) alone, it is measurable, assessable and verifiable.

The College of Applied Biologists (the College) uses a competency framework of seven (7) Professional Practice Competencies (figure 1.0) each with their own key indicators, and a competence continuum (figure 2.0) to align registrants' skills, capabilities, knowledge and competence with their professional practice. The intent of the competency framework, key indicators and competency continuum is to provide the College (the regulator), registrants, employers and the public with clear and consistent expectations and an understanding of what is required of an applied biology professional that is registered and certified by the College.

The Competency Framework, its Key Indicators and the Competence Continuum:

- ◆ Were developed to ensure that College registrants meet and maintain high level practice and competence standards to uphold and protect the public interest;
- ◆ Encompass the required credentials through the registration process to become a College registrant ; and
- ◆ Ensure registrants maintain and enhance the required credentials through the Continued Education, Audit and Practice Review Programs.

Figure 1.0 Professional Practice Competencies.



The College uses the following key indicators (KIs) in each Professional Practice category to assess and verify that a professional meets the minimum competency level requirement: Competent (Figure 2.0). College applicants and registrants are required to demonstrate and maintain competence in at least one (1) key indicator in each Professional Practice Competency category during the Registration, Audit and Practice Review processes.

Records/Data Management & Informatics



1. Demonstrate knowledge of records management and information.
2. Implement policies and protocols relating to records management and informatics in their practice.
3. Respect for confidentiality in their practice.

Professional Accountability



1. Demonstrate an awareness of scope of practice and limitations.
2. Take responsibility for their work and work of others' (if applicable).
3. Exercise skills in applied biology and other professional skills as required for work to be undertaken.

Communication



1. Communicate written and oral information in a clear, concise, and comprehensible manner.
2. Communicate and acknowledge the value of alternative views and hypotheses.
3. Demonstrate the ability to communicate effectively with specialist and non-specialist audiences, colleagues, clients and others.

Scientific knowledge & Concepts



1. Demonstrate knowledge of scientific information and concepts.
2. Incorporate relevant scientific knowledge and concepts in their practice.
3. Use knowledge and scientific understanding to improve the application of existing and emerging scientific concepts, knowledge or technology in their practice.

Laws, regulations & policy



1. Incorporate regulatory and policy requirements in their practice.
2. Implement regulatory and policy requirements in their practice.
3. Demonstrate knowledge of: a) regulatory and policy requirements as a regulated professional; b) Demonstrate knowledge relevant to their practice.

Standards & Practices



1. Demonstrate knowledge of standards and practices relevant to their area of practice.
2. Incorporate standards and practices in their practice.
3. Implement standards and practices in their practice.

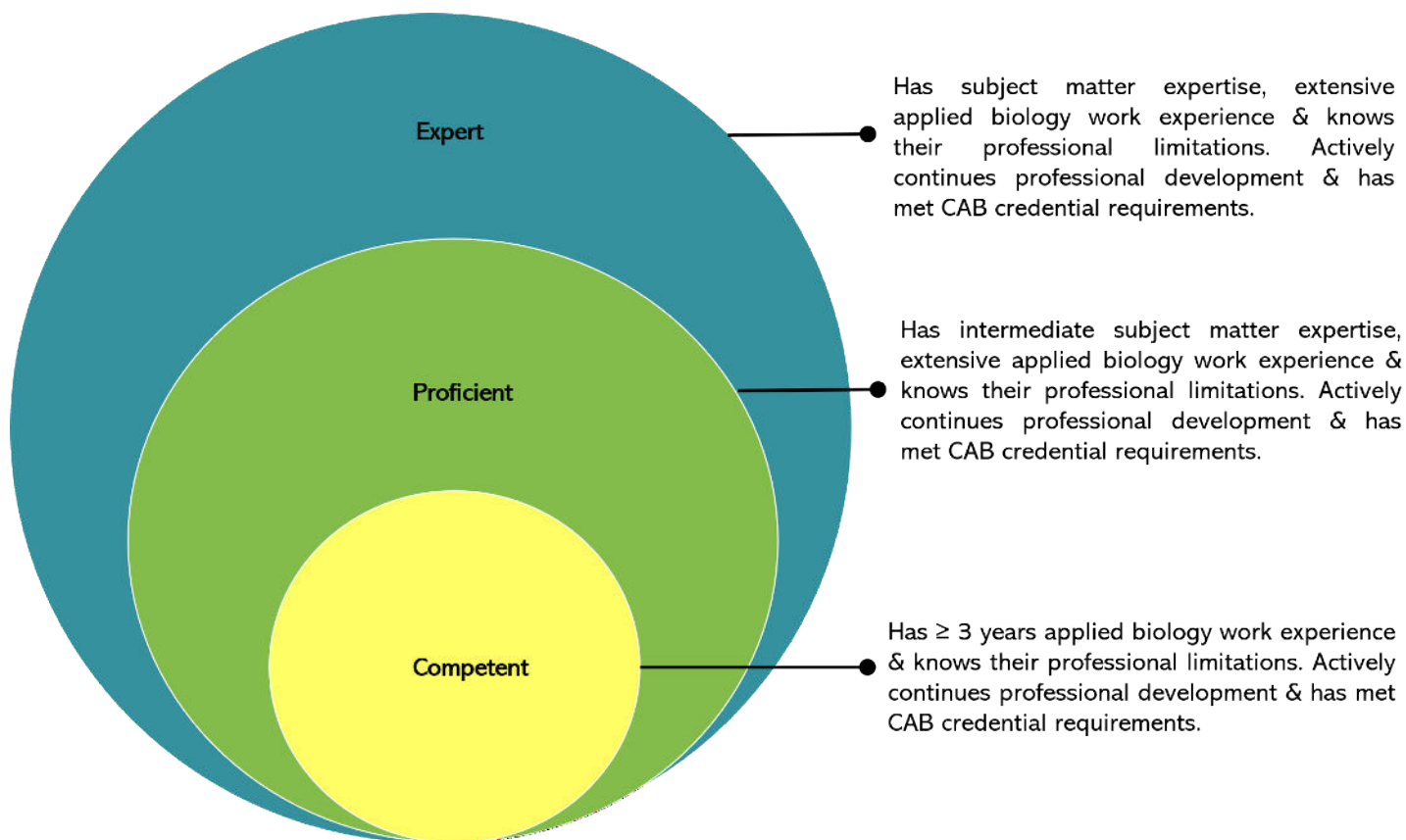
Project &/or Work product Management



1. Provide advice, options /solutions, requirements.
2. Demonstrate skills in evaluation of and in drawing conclusions.
3. Demonstrate skills in the scoping, planning and implementing of projects/assignments.

A registrants Professional Practice Competencies and associated competence level (competent, proficient, expert) (Figure 2.0) are acquired through knowledge (education & training), skills and abilities (work experience, professional practice experience) in each of the Professional Practice Competency categories. The two are linked together.

2.0 Competency Continuum.



During the registration process **an applicant is required to:**

- ♦ Be respectful and professional while interacting with College Staff and employees throughout the process;
- ♦ Demonstrate they meet at least one (1) key indicator in each Professional Practice Competency category;
- ♦ Establish they have a minimum competence level of Competent in each competency category; and
- ♦ Provide References who can verify they meet the Professional Practice Competency categories and are competent.

The College:

- ♦ Assesses the application using the key indicators and competence level requirements;
- ♦ Verifies the information provided by the applicant through information provided by references; and
- ♦ Determines if the applicant is accepted or not accepted. If not accepted, the College provides information regarding any deficiency(ies) and associated resolution(s).

A Registrant is required to:

- ♦ Self-declare their area of expertise and associated competence level with each Professional Practice Competency category;
- ♦ Continue to demonstrate they maintain competence with at least one (1) key indicator in each Professional Practice Competency category through the Audit and Practice Review Processes; and
- ♦ Maintain or enhance their competence and area(s) of expertise through the Continuing Education Program.

The College:

- ♦ Assesses and verifies a Registrants Professional Practice Competencies through the Audit and Practice Review processes, using the key indicators, competent level requirements and Continuing Education activities.

APPENDIX B – Course Requirements and Definitions

The following table summarizes the courses required for each registrant category and associated streams that require specific academic courses.

Academic Requirements	ABT/ABT in Training Streams 1 & 2	RBTech/RBTech in Training Streams 1, 2 & 3	RPBio/Biologist in Training (BIT) - Streams 1 & 2	RPBio Streams 3 & 4	
Communication	Not Applicable	Yes	Yes	Not Applicable	
Numeracy 1 st year or higher (mathematics /statistics)		Yes	Yes		
Applied Biology and Management 2 nd year or higher		Yes	Yes		
Science Courses		Will accept to meet the 20 post-secondary course requirements	Yes - 25	Yes - 25	
Biology Courses			Yes - 13	Yes - 13	
Chemistry 1 st year or higher		No - will accept to meet the 20 post-secondary course requirements	Yes	Not Applicable	
Ecology 2 nd year or higher			Yes		
Statistics 2 nd year or higher			Yes		
Environmental Science		Yes	Will accept to meet the science or biology course requirements		
Field/Laboratory Techniques		Yes			
Natural resource survey skills	Yes	Will accept to meet the 20 post-secondary course requirements			
Applied biology identification skills	Yes				
Field navigation	Yes				
3 of the Following 5 2 nd year or higher Biology courses					
1. Genetics	Not Applicable	Will accept to meet the 20 post-secondary course requirements			Yes
2. Cellular					Yes
3. Systematics or Classification					Yes
4. Physiology					Yes
5. Evolution					Yes

Course Definitions

The following course definitions are provided to assist applicants to understand the College's academic requirements.

Biology courses (general)

Biology courses must include a majority (>50%) of biological concepts, theory, or practice, from a range of disciplines such as geography, environmental science, forestry and forest sciences, ecological restoration, and environmental engineering.

Science courses (general)

Science courses must include a majority (>50%) of scientific concepts, theory, or practice, and may include subject areas such as physics, chemistry, mathematics, geology, geography, environmental science, forestry and forest sciences, ecological restoration, and engineering.

Applied Biology and Management

A course in applied biology will focus on the application of biological, ecological or socioeconomic principles, including law and governance, to the management or conservation of biological resources, elements or systems. The course can focus on a specific group of organisms or consider broader ecosystem-level issues, however, the majority of the course content (i.e., >80%) must consider biological resources, elements or systems, not topics related to the management or conservation of abiotic resources or the more general idea of environmental sustainability.

Courses that typically meet the requirements for this subject category include Conservation Biology, Environmental Biology, Wildlife Management, Fisheries Management, Range Management, Natural Resource Policy, or Landscape Ecology.

The following list represents the general range of topics of Applied Biology and Management:

- Discussion of the application of biological and ecological principles on the use, management or conservation of biological resources
- Discussion of the management or conservation of biological resources
- Discussion of law or governance with direct application to biological resources
- Discussion of the principles of sustainability in the context of biological resources

Applied biology identification skills

Courses or a course module should cover identification of flora and fauna such as plants, fish, mammals, birds, insects and the differences between groups (e.g., trees, shrubs, mosses). Course or course module can also include collection, labeling and preservation of a specimen(s).

Cellular

A cellular course should include cellular chemistry, bioenergetics, enzyme production and function, membranes and cell signaling, membrane transport processes, signal transduction mechanisms, extracellular structures (adhesions, junctions, etc.), chemotrophic energy metabolism, intracellular compartments, phototrophic metabolism, structural basis of cellular information, sexual reproduction, gene expression – transcription and protein synthesis, regulation of gene expression, cytoskeletal

systems, as well as motility and contractility. Courses that would meet this requirement include Cell Biology, Molecular Biology, and Biochemistry.

The following list represents the general range of topics of Cellular biology.

- Cell chemistry, bioenergetics, enzyme structure and function
- Membrane transport
- Membranes and signal transduction mechanisms
- Extracellular structures (adhesions, junctions, etc.)
- Intracellular compartments
- Chemotrophic energy metabolism, phototrophic metabolism, structural basis of cellular information
- Nucleic acid structure and function – replication, transcription, translation and regulation of gene expression
- Cytoskeleton systems, motility and contractility

Chemistry

Chemistry courses can be survey-level first year courses. Courses in chemistry should cover classification of matter, periodic properties of elements, atomic and molecular structure, stoichiometry, chemical reactions, thermochemistry, chemical bonding and an introduction to organic chemistry. Completion of the lab component is not mandatory.

Communications

First year or higher courses that focus on communication skills may include technical writing, scientific writing, or English composition. The aim of this requirement is to ensure the applicant has the ability to provide clear and concise communications which can be interpreted by people at all levels of a decision-making process.

More specifically the graduating student should demonstrate competence in some or all of the following topic areas:

- Development of skills in research and writing academic essays or reports
- Technical writing skills including memos, briefing notes, technical reports and bulletins
- Development of grammatical skills, organizational skills and appropriate use of vocabulary
- Prepare and deliver oral presentations

Ecology

A course in ecology will consider the relationships between living organisms and the abiotic systems within which they occur. Courses focused on ecology can consider a broad range of content from the description and distribution of biomes at the ecosystem level to the factors and theory underlying population and community dynamics. Courses in specific areas of ecology such as population, community or microbial ecology are acceptable as are courses that take a more general perspective on ecology or ecosystem-level processes.

The following list represents the general range of topics of Ecology.

- The relationships between living organisms and the abiotic systems within which they occur
- Description and distribution of biomes at the ecosystem level
- Understanding of the factors and theory underlying population and community dynamics

Evolution

Courses in evolution will cover the contemporary theory of evolution, such as variation, descent, natural selection, adaptation, speciation, and extinction on both micro- and macro-evolutionary scales. Topics include origins of living systems; species and their origins and extinctions; adaptation and constraints; systematics; evolutionary ethics.

Examples of courses that may meet these criteria include but are not limited to:

- Evolution
- Diversity and Evolution
- Evolutionary Genetics
- Ecology and Evolution

Environmental Science

A course in environmental science will discuss multi-disciplinary topics related to natural resource use and sustainability. The course may be as specific as discussing one resource, to the broader global scale of environmental sustainability.

Courses that typically meet the requirements for this subject category include Fish and wildlife ecology, environmental legislation and policy, earth science, soils, hydrology, climatology.

The following list represents the general range of topics of Environmental Science.

- Environmental sustainability
- Natural resource management
- Ecological restoration
- Legislation and policy

Field/Laboratory Techniques

A course in field/laboratory techniques will focus on the skills and techniques required either in the field or laboratory (lab) for the practice of applied biology as defined in the [Applied Biologists Regulation](#). There should be a focus on following standardized data collection methodology.

Courses that typically meet the requirements for this subject category include chemistry with a lab component, biology with a lab component, field studies, surveying, maps and navigation, fish or plant identification.

The following list represents the general range of topics of Field/Laboratory Techniques.

- Field survey methods
- Laboratory methods
- GPS data collection
- Ecosystem classification

Field Navigation

Courses or a course module should cover navigation skills required for field work. This may include the use of and interpretation of (hardcopy and digital) maps, compasses, GPS systems (handheld and software), hip chain, navigation data collection and storage.

Genetics

Courses in genetics should cover structure and function of genes, chromosomes and genomes, biological variation resulting from recombination, mutation, and selection, population genetics, use of genetic methods to analyze protein function, gene regulation and inherited disease.

Examples of courses that meet these criteria include Genetics, Molecular Genetics, or Genomics.

Natural Resource Survey Skills

Courses or a course module should cover data collection (e.g., hard copy and/or digital), proper field note taking, common field equipment requirements, use, maintenance and care (e.g., calibration, storage), quality assurance and data integrity and security.

Numeracy (Mathematics)

Mathematics courses need to be at a first year or higher university level. Introductory courses in calculus, applied calculus, finite math, linear algebra or a first year statistics course will generally meet this requirement. Technical math courses may not meet the requirement unless they transfer to an equivalent first year math course at a college or university through a recognized transfer agreement.

Numeracy (Statistics)

A course in statistics will include applications of parametric and non-parametric statistical methods.

The following list represents the general range of topics found in a Statistics course. It is not expected that all topics will be addressed during a major or degree. However, it is expected that graduating students will engage with enough topics to gain fundamental knowledge necessary to be competent in this area.

- Descriptive statistics and probability
- Types of distributions
- Hypothesis testing on means and proportions
- Experimental and sampling design
- Analysis of variance, regression and correlation
- Understand or conduct survival analyses

Physiology

A course in physiology will focus on the relationship between the structure and functioning of individual organisms relative to the environments they occupy. The majority of course content (i.e. >80%) should consider the physiology of the organism in the context of environmental responses and/or the interaction between anatomical structure and life history including adaptive behaviour. The course can include the study of a single taxonomic group or a broader perspective across a number of Kingdoms.

The following table represents the general range of topics of cell, animal, or plant physiology courses.

<u>Cell Physiology Course</u>	<u>Animal Physiology Course (including human physiology)</u>	<u>Plant Physiology Course</u>
• Structure and function of the cytoskeleton • Structure and function of the cell membrane • Structure and function of cell organelles • Topics on cell dynamics and	• Principles of homeostasis • Discussions of cardiovascular systems • Discussions of respiratory system • Discussion of osmoregulatory system	• Mechanisms and regulation of functional processes contributing to assimilation, transport and utilization of water, mineral nutrients and carbon

		Understanding of cell
bioenergetics Topics on regulation of cellular activities	- Discussions of endocrine system - Discussions of excitable membranes of nerve and muscle - Discussions of reproduction	division, sexual and asexual reproduction - Understanding of differentiation and functions of various tissue types - Understanding of the action of major growth regulators and photomorphogenesis

Systematics or Classification

A course in systematics or classification will focus on the nomenclature, identification, and categorization of organisms from within one or more taxonomic groups. Course content often includes a description of the unique anatomy or other differentiating characteristics of the organisms under study. Courses in Systematics or Classification may require students to develop a collection of specimens (e.g. herbarium) or develop the skills to identify individual species or genera from a broader set of related organisms.

The following list represents the general range of topics of systematics or classification.

- Nomenclature, identification and categorization within one or more taxonomic group
- Description of unique anatomy or other differing characteristics
- Development of a collection of specimens or development of skills to identify species

Examples of courses that typically meet this subject category include:

- Invertebrate Zoology
- Vertebrate Zoology
- Vascular Plants
- Non-Vascular Plants
- Ornithology
- Herpetology
- Mammalogy
- Phylogenetics
- Forest Classification and Silvics

APPENDIX C – Registrant Application Checklists

Applied Biology Technician (ABT) and ABT in Training Streams 1 and 2 Checklist

Applied Biology Technician (ABT) and ABT in Training		
Requirement:	ABT Streams 1 & 2 Y/N	ABT in Training Streams 1 & 2 Y/N
Education/Training		
≥ 200 hours or 5 instructional weeks of course work with or without a certificate		
Including courses or course modules in each of the following areas:		
Natural resource survey skills		
Applied biology identification skills		
Field navigation		
Experience - meets the practice of applied biology (Applied Biologists Regulation)		Not Applicable
12 months experience in applied biology in the 10 years prior to application		
Professional Practice Competencies		
Meets all 7 Professional Practice Competencies		
1 Reference to support Professional Practice Competencies		

Applied Biology Technician (ABT) Stream 3 Checklist

Applied Biology Technician (ABT)	
Requirement	Stream 3 Y/N
Education/Training	
36 months (≥3 years) experience in applied biology in the last years prior to application	
Professional Practice Competencies	
Meets all 7 Professional Practice Competencies	
2 References to support Professional Practice Competencies	

Registered Biology Technologist (RBTech) and RBTech in Training Streams 1 and 2 Checklist

Registered Biology Technologist (RBTech) and RBTech in Training Requirements	RBTech Streams 1 & 2 Y/N	RBTech in Training Streams 1 & 2 Y/N
Education/Training		
Diploma		
Required Courses:		
Communications		
Mathematics/Statistics (Numeracy)		
Environmental Science		
Field/Laboratory Techniques		
Applied Biology and Management		
Experience - meets the practice of applied biology (Applied Biologists Regulation)		Not Applicable
≥ 24 months (≥ 2 years) experience in applied biology in the 10 years prior to application		
Professional Work Products (PWPs)		
1. Do your PWPs meet 1 of the key indicators for each of the following Professional Practice Competencies		
• Scientific Knowledge		
• Laws, Regulations and Policies		
• Standards & Practices		
• Communication		
And include the following:		
• Methods		
• Data analysis within established standards, parameters &/or guidelines		
• Conclusion(s) and/or Recommendations		
• Communicates technical scientific information to a non-expert audience.		
2. Describe the purpose/intent/background of each of the professional work product(s) submitted		
Professional Practice Competencies		
Meets all 7 Professional Practice Competencies		
2 References to support Professional Practice Competencies		

Registered Biology Technologist (RBTech) Stream 3 Checklist

Registered Biology Technologist (RBTech) Requirement	Stream 3 Y/N
Education/Training	
≥20 Post Secondary Courses	
Required Courses:	
Communications	
Mathematics/Statistics (Numeracy)	
Environmental Science	
Field/Laboratory Techniques	
Applied Biology and Management	
Experience - meets the practice of applied biology (Applied Biologists Regulation)	
24 months' experience in applied biology in the 10 years prior to application	
Professional Work Products (PWPs)	
1. Do your PWPs meet 1 of the key indicators for each of the following Professional Practice Competences	
• Scientific Knowledge	
• Laws, Regulations and Policies	
• Standards & Practices	
• Communication	
And include the following:	
• Methods	
• Data analysis within established standards, parameters &/or guidelines	
• Conclusion(s) and/or Recommendations	
• Communicates technical scientific information to a non-expert audience.	
2. Describe the purpose/intent/background of each of the professional work product(s) submitted	
Professional Practice Competencies	
Meets all 7 Professional Practice Competencies	
2 References to support Professional Practice Competencies	

**Registered Professional Biologist (RPBio) and Biologist in Training (BIT)
Streams 1 and 2 Checklist**

Registered Professional Biologist (RPBio) and Biologist in Training (BIT) Requirements	RPBio Streams 1 and 2 Y/N	BIT Streams 1 and 2 Y/N
Education/Training		
Bachelor's or Master's degree		
25 Science Courses		
13 Biology Courses		
Core Course Requirements		
Communications (1 st year or higher)		
Chemistry (1 st year or higher)		
Numeracy (1 st year or higher)		
Statistics (2 nd year or higher)		
Applied Biology or Management (2 nd year or higher)		
Ecology (2 nd year or higher)		
Do you have 3 of the 5 following courses (2nd year or higher):		
Genetics		
Cellular		
Systematics		
Physiology		
Evolution		
Experience - meets the practice of applied biology (Applied Biologists Regulation)		Not Applicable
36 months' experience in the 10 years prior to application		
Professional Work Products (PWP's)		
1. Do your PWP's meet 1 of the key indicators for each of the following Professional Practice Competencies		
• Scientific Knowledge		
• Laws, Regulations and Policies		
• Standards & Practices		
• Communication		
And include the following:		
• Methods		
• Data analysis		
• Conclusion(s) and/or Recommendations		
• Communicates technical scientific information to a non-expert audience.		
2. Describe the purpose/intent/background of each of the professional work product(s) submitted		
Professional Practice Competencies		
Meets all 7 Professional Practice Competencies		
3 References to support Professional Practice Competencies		

Registered Professional Biologist (RPBio) Stream 3 Checklist

Registered Professional Biologist (RPBio) Requirement	Stream 3 Y/N
Education/Training	
Bachelor's or Master's degree	
25 Science Courses	
13 Biology Courses	
Experience - meets the practice of applied biology (Applied Biologists Regulation)	
Degree plus ≥ 15 years (180 months) of applied biology experience	
Professional Work Products (PWP)	
1. Do your PWP meet 1 of the key indicators for each of the following Professional Practice Competencies	
• Scientific Knowledge	
• Laws, Regulations and Policies	
• Standards & Practices	
• Communication	
And include the following:	
• Methods	
• Data analysis	
• Conclusion(s) and/or Recommendations	
• Communicates technical scientific information to a non-expert audience.	
2. Describe the purpose/intent/background of each of the professional work product(s) submitted	
Professional Practice Competencies	
Meets all 7 Professional Practice Competencies	
3 References to support Professional Practice Competencies	

Registered Professional Biologist (RPBio) Stream 4 Checklist

Registered Professional Biologist (RPBio) Requirement:	Streams 3 & 4
Education/Training	
Master of Science (MSc) &/or Doctor of Philosophy (PhD) in an applied biology topic	
25 Science Courses	
13 Biology Courses	
Experience - meets the practice of applied biology (Applied Biologists Regulation)	
MSc plus 13 years (156 months) of applied biology experience	
PhD plus 11 years (122 months) of applied biology experience	
Professional Work Products (PWP)	
1. Do your PWP meet 1 of the key indicators for each of the following Professional Practice Competences	
meet 1 of the key indicators for each of the following Professional Practice Competences	
Scientific Knowledge	
Laws, Regulations and Policies	
Standards & Practices	
Communication	
And include the following:	
Methods	
Data analysis	
Conclusion(s) and/or Recommendations	
Communicates technical scientific information to a non-expert audience.	
2. Describe the purpose/intent/background of each of the professional work product(s) submitted	
Professional Practice Competencies	
Meets all 7 Professional Practice Competencies	
3 References to support Professional Practice Competencies	

Applied Biology – Limited Licensee Application Checklist

Applied Biology – Limited Licensee Requirements	Y/N
Education/Training	
Proof of education that supports the license being applied for	
Experience	
≥ 5 years (≥ 60 months) of applied biology experience as defined in the Applied Biologists Regulation with 2 years in the specific area of the license being applied for	
Professional Work Products (PWPs)	
1. Do your PWPs meet 1 of the key indicators for each of the following Professional Practice Competencies	
• Scientific Knowledge	
• Laws, Regulations and Policies	
• Standards & Practices	
• Communication	
And include the following:	
• Methods	
• Data analysis within established standards, parameters &/or guidelines	
• Conclusion(s) and/or Recommendations	
• Communicates technical scientific information to a non-expert audience.	
2. Describe the purpose/intent/background of each of the professional work product(s) submitted	
Professional Practice Competencies	
Meets all 7 Professional Practice Competencies	
3 References to support Professional Practice Competencies	